

Power Your Mobile Applications with Firebase

Firebase is a Google mobile platform that helps you quickly develop high quality apps and thereby scale your business. This article covers the range of services it offers to the user.



Firebase started off as a real-time database in 2011 and saw some traction in key applications. The real upswing for the platform came about when Google acquired it in 2014, and since then a significant number of features have been developed for this database. It has now become a complete platform that provides services for development, testing, distribution, analytics and more.

Firebase categorises its services into two areas that help you do the following:

- Develop and test your application
- Grow and engage your audience

Multiple services are then provided under each of these areas. You may or may not use all of them but Google has done a good job of explaining the scenarios for which you will need these services. In fact, years of mobile application development experience, coupled with Google engaging with the growing number of mobile users, has resulted in commonly used application development and growth strategies along with the tools that are needed to make a mobile application successful.

Let us look at the two categories of services and see what Firebase has to offer.

Develop and test your application

The key thing, initially, is to develop your mobile application. For this, Firebase provides multiple services

and its key module here is the Firebase Realtime Database. This database is fully managed by Google and is akin to a Database-as-a-Service model. It provides SDKs on popular mobile platforms like Android and iOS, and it's a breeze to integrate the functionality into your application. The Firebase database allows multiple users to update it and also provides synchronisation across multiple users.

Now, let us understand what more we need from a mobile development perspective. Here are some of the requirements, other than the core business functionality that you are going to build into your application:

- 1) Collaboration across devices
- 2) Centralised crash reporting in case of mobile application errors
- 3) Authentication via multiple Auth providers
- 4) Use of optional cloud storage and long running back-end processes for some of your mobile app's functionality
- 5) Test your app on multiple phone models before release
- 6) Performance monitoring of your app while it is in the field

Each of the above requirements has a matching service that is provided as part of the Firebase platform, and these are covered below.

Firebase Realtime Database: As mentioned earlier, this is a fully managed real-time database that helps in collaboration by sharing the updates across all the devices that use the database. You can integrate the mobile SDKs into